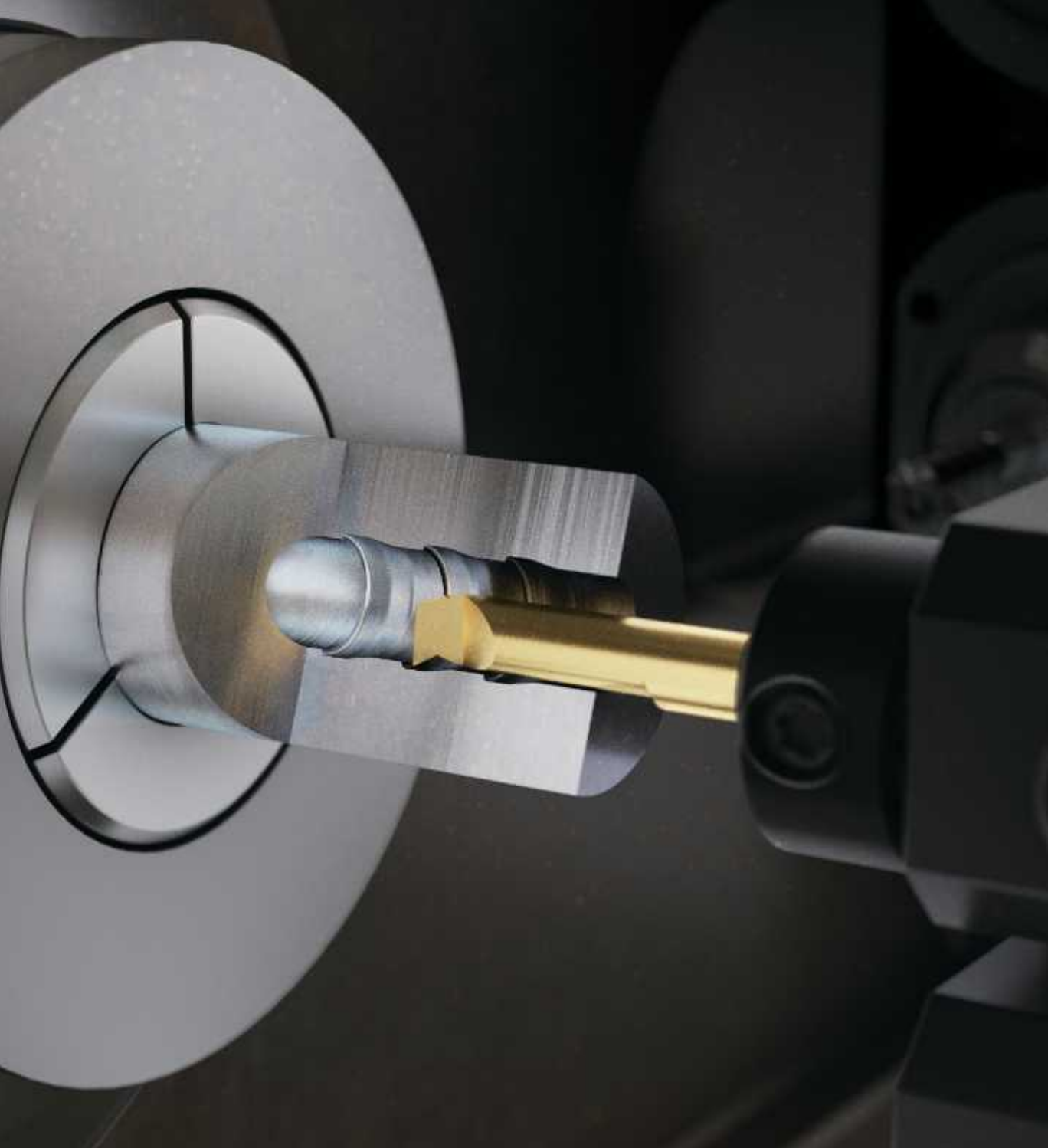


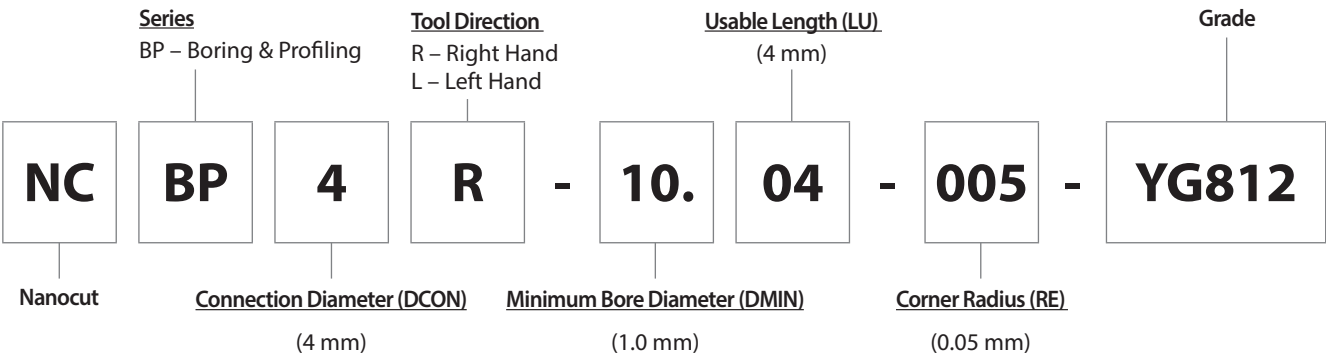


SOLID MINIATURE TURNING for SMALL BORE

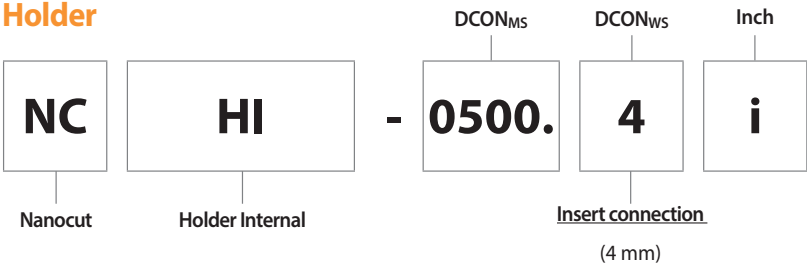


Code Keys - Boring Tool & Holder

Boring Tool



Holder



Success Story

Cutting Conditions

Size	Ø.236", R.008"
Work Material	4140 (Alloy Steel) (HRC 20)
Vc	328 ft/min
N (RPM)	637 rev/min
Fn	.004 in/rev
Ap	Axial: .390" Radial: .006"
Coolant	Wet Cut
Operation	Boring
Machine	Turn Mill

Less is
better



Edge wear (after 30 min.)

YG



Competitor



Overview

At present, as demand for small parts (camera lens, mobile phone parts) and medical instruments (implants) increases, demand for small-diameter products capable of high-precision processing is increasing.

Application

- Turning of Small Bore Components
- Internal Turning(Boring), Grooving and Threading

Features

- Minimum Diameter(Boring & Profiling series) : Ø 1 mm (.039")
- Internal Coolant for Longer Tool life and Enhanced Chip Evacuation
- Secure Connection Design: Pin + Slant Positioning
- 10 Geometries for Various Applications

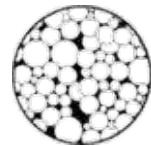
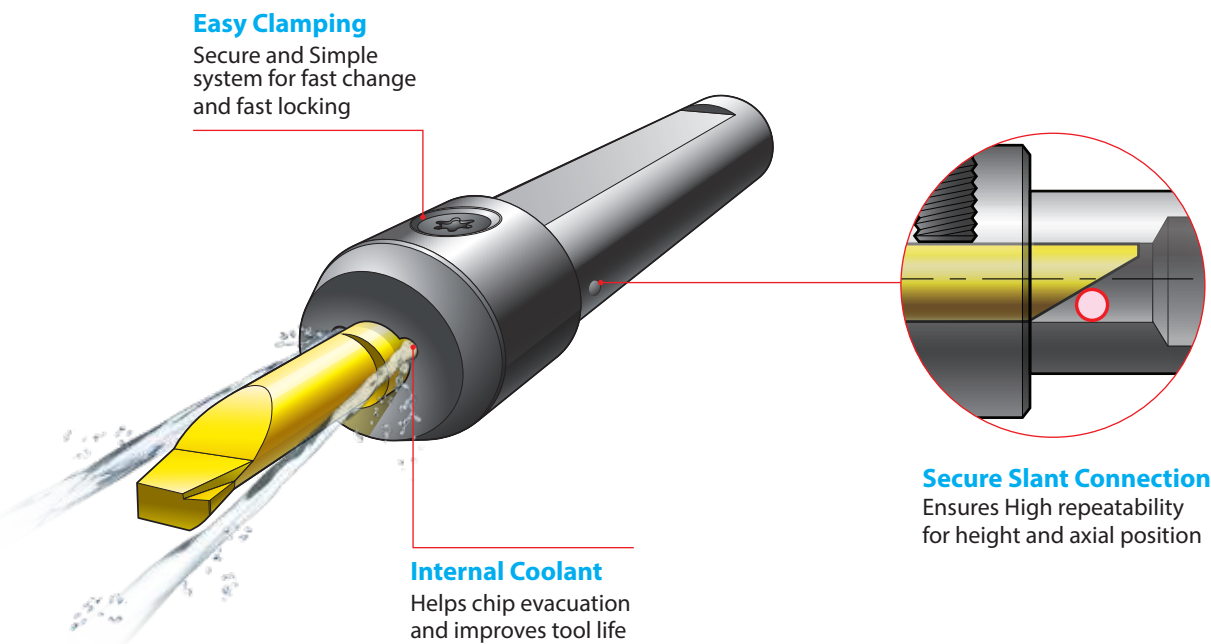
Benefits

- Reduced Machine Down Time
- Lower Machining Cost

YG812 - Micro Grain Carbide Grade

P10 - P20 **M20 - M30** Submicron Grade Carbide Substrate Material for high toughness
K20 - K30 **S10 - S25** and wear resistance realizes high precision machining

Advantages of Nanocut

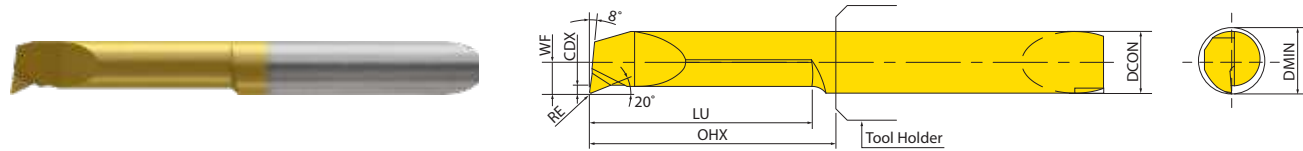


Overview

Recommended Cutting Conditions : p.191

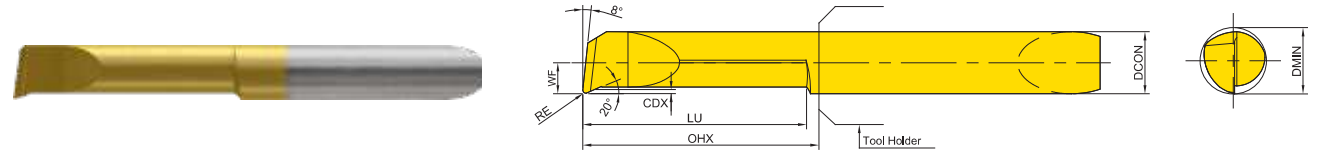
Series		KAPR (°)	DCON mm(inch)	Page
BP Boring & Profiling		98	4 (.157) 6 (.236)	104
		98		105
		90		106
		98		107
Series		CW (inch)	DCON mm(inch)	Page
GR Grooving Round		.039 - .079	4 (.157) 6 (.236)	108
		.031 - .079		109
FI Face Grooving Internal		.039 - .118		110
FE Face Grooving External		.039 - .118		110
Series		WF (inch)	DCON mm(inch)	Page
CH Chamfering		.020 - .043	4 (.157) 6 (.236)	111
		.079 - .116		112
Series		DCON _{WS} (inch)	DCON _{MS} (inch)	Page
Holder		4 (.157) 6 (.236)	.500 - 1.000	113

Boring Tool
BP - Boring & Profiling



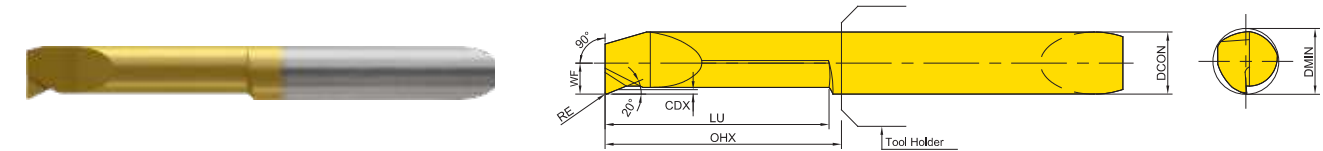
KAPR	DCON mm(inch)	Designation	EDP	Inch ⁽ⁱⁿ⁾							YG812
				DMIN	LU	RE	CDX	WF	OHX	OAL	
98°	4 (.157)	NCBP4R - 10.04 - 005 - YG812	NCBP04R1004005T	.039	.16	.002	.004	.018	.51	1.10	●
		NCBP4R - 10.04 - 010 - YG812	NCBP04R1004010T	.039	.16	.004	.004	.018	.51	1.10	●
		NCBP4R - 10.06 - 005 - YG812	NCBP04R1006005T	.039	.24	.002	.004	.018	.51	1.10	●
		NCBP4R - 10.06 - 010 - YG812	NCBP04R1006010T	.039	.24	.004	.004	.018	.51	1.10	●
		NCBP4R - 17.06 - 005 - YG812	NCBP04R1706005T	.067	.24	.002	.008	.028	.51	1.10	○
		NCBP4R - 17.06 - 010 - YG812	NCBP04R1706010T	.067	.24	.004	.008	.028	.51	1.10	●
		NCBP4R - 17.09 - 005 - YG812	NCBP04R1709005T	.067	.35	.002	.008	.028	.51	1.10	○
		NCBP4R - 17.09 - 010 - YG812	NCBP04R1709010T	.067	.35	.004	.008	.028	.51	1.10	●
		NCBP4R - 22.06 - 005 - YG812	NCBP04R2206005T	.087	.24	.002	.008	.037	.51	1.10	○
		NCBP4R - 22.06 - 010 - YG812	NCBP04R2206010T	.087	.24	.004	.008	.037	.51	1.10	●
		NCBP4R - 22.09 - 005 - YG812	NCBP04R2209005T	.087	.35	.002	.008	.037	.51	1.10	○
		NCBP4R - 22.09 - 010 - YG812	NCBP04R2209010T	.087	.35	.004	.008	.037	.51	1.10	●
		NCBP4R - 22.13 - 010 - YG812	NCBP04R2213010T	.087	.51	.004	.008	.037	.71	1.30	●
		NCBP4R - 27.10 - 005 - YG812	NCBP04R2710005T	.106	.39	.002	.008	.047	.51	1.10	○
		NCBP4R - 27.10 - 015 - YG812	NCBP04R2710015T	.106	.39	.006	.008	.047	.51	1.10	●
		NCBP4R - 27.15 - 005 - YG812	NCBP04R2715005T	.106	.59	.002	.008	.047	.71	1.30	○
		NCBP4R - 27.15 - 015 - YG812	NCBP04R2715015T	.106	.59	.006	.008	.047	.71	1.30	●
		NCBP4R - 32.10 - 015 - YG812	NCBP04R3210015T	.126	.39	.006	.008	.057	.51	1.10	●
		NCBP4R - 32.15 - 015 - YG812	NCBP04R3215015T	.126	.59	.006	.008	.057	.71	1.30	●
		NCBP4R - 32.20 - 005 - YG812	NCBP04R3220005T	.126	.79	.002	.008	.057	.91	1.50	○
		NCBP4R - 32.20 - 015 - YG812	NCBP04R3220015T	.126	.79	.006	.008	.057	.91	1.50	●
		NCBP4R - 42.10 - 015 - YG812	NCBP04R4210015T	.165	.39	.006	.012	.077	.51	1.10	●
		NCBP4R - 42.15 - 005 - YG812	NCBP04R4215005T	.165	.59	.002	.012	.077	.71	1.30	○
		NCBP4R - 42.15 - 015 - YG812	NCBP04R4215015T	.165	.59	.006	.012	.077	.71	1.30	●
		NCBP4R - 42.20 - 005 - YG812	NCBP04R4220005T	.165	.79	.002	.012	.077	.91	1.50	○
		NCBP4R - 42.20 - 015 - YG812	NCBP04R4220015T	.165	.79	.006	.012	.077	.91	1.50	●
		NCBP4R - 42.25 - 005 - YG812	NCBP04R4225005T	.165	.98	.002	.012	.077	1.10	1.69	○
		NCBP4R - 42.25 - 015 - YG812	NCBP04R4225015T	.165	.98	.006	.012	.077	1.10	1.69	●
	6 (.236)	NCBP6R - 62.15 - 020 - YG812	NCBP06R6215020T	.244	.59	.008	.020	.116	.71	1.69	●
		NCBP6R - 62.20 - 020 - YG812	NCBP06R6220020T	.244	.79	.008	.020	.116	.91	1.89	●
		NCBP6R - 62.25 - 020 - YG812	NCBP06R6225020T	.244	.98	.008	.020	.116	1.10	2.09	●
		NCBP6R - 62.30 - 020 - YG812	NCBP06R6230020T	.244	1.18	.008	.020	.116	1.30	2.28	●
		NCBP6R - 62.35 - 020 - YG812	NCBP06R6235020T	.244	1.38	.008	.020	.116	1.50	2.48	●
		NCBP6R - 62.40 - 020 - YG812	NCBP06R6240020T	.244	1.57	.008	.020	.116	1.69	2.68	●

Boring Tool
BO - Boring with Chip Breaker



KAPR	DCON mm(inch)	Designation	EDP	Inch ⁽ⁱⁿ⁾							YG812
				DMIN	LU	RE	CDX	WF	OHX	OAL	
98°	4 (.157)	NCBO4R - 10.06 - 005 - YG812	NCBO04R1006005T	.039	.24	.002	.004	.018	.51	1.10	○
		NCBO4R - 10.06 - 010 - YG812	NCBO04R1006010T	.039	.24	.004	.004	.018	.51	1.10	●
		NCBO4R - 17.09 - 005 - YG812	NCBO04R1709005T	.067	.35	.002	.008	.028	.51	1.10	○
		NCBO4R - 17.09 - 010 - YG812	NCBO04R1709010T	.067	.35	.004	.008	.028	.51	1.10	○
		NCBO4R - 22.09 - 005 - YG812	NCBO04R2209005T	.087	.35	.002	.008	.037	.51	1.10	○
		NCBO4R - 22.09 - 010 - YG812	NCBO04R2209010T	.087	.35	.004	.008	.037	.51	1.10	●
		NCBO4R - 22.13 - 010 - YG812	NCBO04R2213010T	.087	.51	.004	.008	.037	.71	1.30	○
		NCBO4R - 27.15 - 005 - YG812	NCBO04R2715005T	.106	.59	.002	.008	.047	.71	1.30	○
		NCBO4R - 27.15 - 015 - YG812	NCBO04R2715015T	.106	.59	.006	.008	.047	.71	1.30	○
		NCBO4R - 32.15 - 015 - YG812	NCBO04R3215015T	.126	.59	.006	.008	.057	.71	1.30	●
		NCBO4R - 32.20 - 005 - YG812	NCBO04R3220005T	.126	.79	.002	.008	.057	.91	1.50	○
		NCBO4R - 32.20 - 015 - YG812	NCBO04R3220015T	.126	.79	.006	.008	.057	.91	1.50	●
		NCBO4R - 42.20 - 005 - YG812	NCBO04R4220005T	.165	.79	.002	.012	.077	.91	1.50	○
		NCBO4R - 42.20 - 015 - YG812	NCBO04R4220015T	.165	.79	.006	.012	.077	.91	1.50	●
		NCBO4R - 42.25 - 005 - YG812	NCBO04R4225005T	.165	.98	.002	.012	.077	1.10	1.69	○
		NCBO4R - 42.25 - 015 - YG812	NCBO04R4225015T	.165	.98	.006	.012	.077	1.10	1.69	●
	6 (.236)	NCBO6R - 62.15 - 020 - YG812	NCBO06R6215020T	.244	.59	.008	.020	.116	.81	1.69	●
		NCBO6R - 62.25 - 020 - YG812	NCBO06R6225020T	.244	.98	.008	.020	.116	1.10	2.09	●
		NCBO6R - 62.30 - 020 - YG812	NCBO06R6230020T	.244	1.18	.008	.020	.116	1.30	2.28	●
		NCBO6R - 62.35 - 020 - YG812	NCBO06R6235020T	.244	1.38	.008	.020	.116	1.50	2.48	●
		NCBO6R - 62.40 - 020 - YG812	NCBO06R6240020T	.244	1.57	.008	.020	.116	1.69	2.68	○

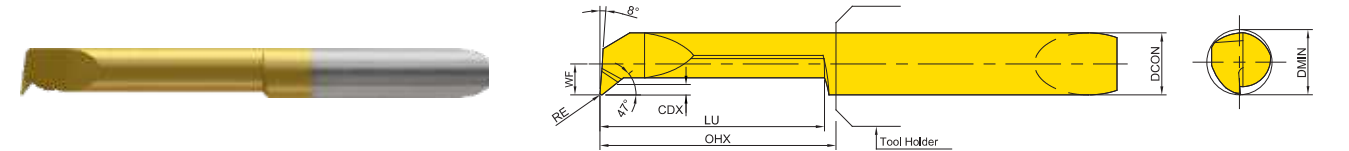
Boring Tool
BF - 90° Boring



●: Stock item
○: Order made item

KAPR	DCON mm(inch)	Designation	EDP	Inch ⁽ⁱⁿ⁾							YG812
				DMIN	LU	RE	CDX	WF	OHX	OAL	
90°	4 (.157)	NCBF4R - 10.04 - 010 - YG812	NCBF04R1004010T	.039	.16	.004	.004	.018	.51	1.10	○
		NCBF4R - 10.06 - 010 - YG812	NCBF04R1006010T	.039	.24	.004	.004	.018	.51	1.10	○
		NCBF4R - 17.06 - 010 - YG812	NCBF04R1706010T	.067	.24	.004	.008	.028	.51	1.10	○
		NCBF4R - 17.09 - 010 - YG812	NCBF04R1709010T	.067	.35	.004	.008	.028	.51	1.10	○
		NCBF4R - 22.06 - 010 - YG812	NCBF04R2206010T	.087	.24	.004	.008	.037	.51	1.10	○
		NCBF4R - 22.09 - 010 - YG812	NCBF04R2209010T	.087	.35	.004	.008	.037	.51	1.10	○
		NCBF4R - 22.13 - 010 - YG812	NCBF04R2213010T	.087	.51	.004	.008	.037	.71	1.30	○
		NCBF4R - 27.10 - 015 - YG812	NCBF04R2710015T	.106	.39	.006	.008	.047	.51	1.10	○
		NCBF4R - 27.15 - 015 - YG812	NCBF04R2715015T	.106	.59	.006	.008	.047	.71	1.30	○
		NCBF4R - 32.10 - 015 - YG812	NCBF04R3210015T	.126	.39	.006	.008	.057	.51	1.10	○
		NCBF4R - 32.15 - 015 - YG812	NCBF04R3215015T	.126	.59	.006	.008	.057	.71	1.30	○
		NCBF4R - 32.20 - 015 - YG812	NCBF04R3220015T	.126	.79	.006	.008	.057	.91	1.50	○
		NCBF4R - 42.10 - 015 - YG812	NCBF04R4210015T	.165	.39	.006	.012	.077	.51	1.10	○
		NCBF4R - 42.15 - 015 - YG812	NCBF04R4215015T	.165	.59	.006	.012	.077	.71	1.30	○
		NCBF4R - 42.20 - 015 - YG812	NCBF04R4220015T	.165	.79	.006	.012	.077	.91	1.50	○
		NCBF4R - 42.25 - 015 - YG812	NCBF04R4225015T	.165	.98	.006	.012	.077	1.10	1.69	○
	6 (.236)	NCBF6R - 62.15 - 020 - YG812	NCBF06R6215020T	.244	.59	.008	.020	.116	.71	1.69	○
		NCBF6R - 62.20 - 020 - YG812	NCBF06R6220020T	.244	.79	.008	.020	.116	.91	1.89	○
		NCBF6R - 62.25 - 020 - YG812	NCBF06R6225020T	.244	.98	.008	.020	.116	1.10	2.09	○
		NCBF6R - 62.30 - 020 - YG812	NCBF06R6230020T	.244	1.18	.008	.020	.116	1.30	2.28	○
		NCBF6R - 62.35 - 020 - YG812	NCBF06R6235020T	.244	1.38	.008	.020	.116	1.50	2.48	○
		NCBF6R - 62.40 - 020 - YG812	NCBF06R6240020T	.244	1.57	.008	.020	.116	1.69	2.68	○

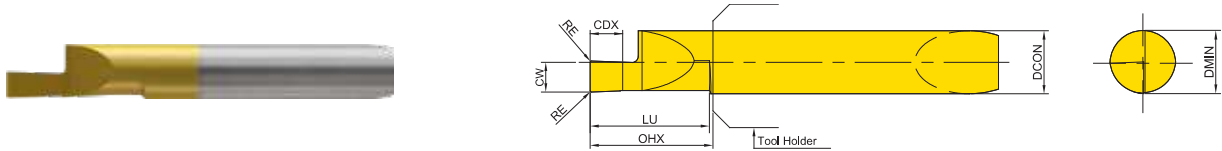
Boring Tool
PR - Profiling



●: Stock item
○: Order made item

KAPR	DCON mm(inch)	Designation	EDP	Inch ⁽ⁱⁿ⁾							YG812
				DMIN	LU	RE	CDX	WF	OHX	OAL	
98°	4 (.157)	NCPR4R - 10.04 - 010 - YG812	NCPR04R1004010T	.039	.16	.004	.004	.018	.51	1.10	○
		NCPR4R - 10.06 - 010 - YG812	NCPR04R1006010T	.039	.24	.004	.004	.018	.51	1.10	○
		NCPR4R - 17.06 - 010 - YG812	NCPR04R1706010T	.067	.24	.004	.008	.028	.51	1.10	○
		NCPR4R - 17.09 - 010 - YG812	NCPR04R1709010T	.067	.35	.004	.008	.028	.51	1.10	○
		NCPR4R - 22.06 - 010 - YG812	NCPR04R2206010T	.087	.24	.004	.016	.037	.51	1.10	○
		NCPR4R - 22.09 - 010 - YG812	NCPR04R2209010T	.087	.35	.004	.016	.037	.51	1.10	○
		NCPR4R - 22.13 - 010 - YG812	NCPR04R2213010T	.087	.51	.004	.016	.037	.71	1.30	○
		NCPR4R - 27.10 - 015 - YG812	NCPR04R2710015T	.106	.39	.006	.020	.047	.51	1.10	○
		NCPR4R - 27.15 - 015 - YG812	NCPR04R2715015T	.106	.59	.006	.020	.047	.71	1.30	○
		NCPR4R - 32.10 - 015 - YG812	NCPR04R3210015T	.126	.39	.006	.024	.057	.51	1.10	○
		NCPR4R - 32.15 - 015 - YG812	NCPR04R3215015T	.126	.59	.006	.024	.057	.71	1.30	○
		NCPR4R - 32.20 - 015 - YG812	NCPR04R3220015T	.126	.79	.006	.024	.057	.91	1.50	○
		NCPR4R - 42.10 - 015 - YG812	NCPR04R4210015T	.165	.39	.006	.031	.077	.51	1.10	●
		NCPR4R - 42.15 - 015 - YG812	NCPR04R4215015T	.165	.59	.006	.031	.077	.71	1.30	●
		NCPR4R - 42.20 - 015 - YG812	NCPR04R4220015T	.165	.79	.006	.031	.077	.91	1.50	●
		NCPR4R - 42.25 - 015 - YG812	NCPR04R4225015T	.165	.98	.006	.031	.077	1.10	1.69	○
	6 (.236)	NCPR6R - 62.15 - 020 - YG812	NCPR06R6215020T	.244	.59	.008	.071	.116	.71	1.69	●
		NCPR6R - 62.20 - 020 - YG812	NCPR06R6220020T	.244	.79	.008	.071	.116	.91	1.89	●
		NCPR6R - 62.25 - 020 - YG812	NCPR06R6225020T	.244	.98	.008	.071	.116	1.10	2.09	○
		NCPR6R - 62.30 - 015 - YG812	NCPR06R6230015T	.244	1.18	.006	.071	.116	1.30	2.28	●
		NCPR6R - 62.30 - 020 - YG812	NCPR06R6230020T	.244	1.18	.008	.071	.116	1.30	2.28	○
		NCPR6R - 62.35 - 020 - YG812	NCPR06R6235020T	.244	1.38	.008	.071	.116	1.50	2.48	○
		NCPR6R - 62.40 - 020 - YG812	NCPR06R6240020T	.244	1.57	.008	.071	.116	1.69	2.68	○

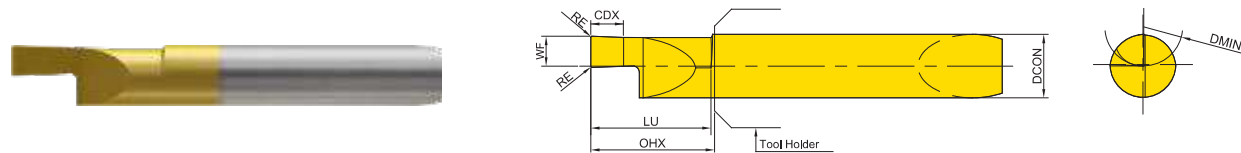
Boring Tool
FI - Face Grooving Internal



●: Stock item
○: Order made item

DCON (mm)	Designation	EDP	Inch ⁽ⁱⁿ⁾								YG812
			DMIN	LU	RE	CW	CDX	WF	OHX	OAL	
6 (.236)	NCFI6R - 62.15 - 10x20 - YG812	NCFI06R6215010T	.244	.59	.004	.039	.079	.116	.71	1.69	●
	NCFI6R - 62.15 - 15x30 - YG812	NCFI06R6215015T	.244	.59	.004	.059	.118	.116	.71	1.69	●
	NCFI6R - 62.15 - 20x40 - YG812	NCFI06R6215020T	.244	.59	.004	.079	.157	.116	.71	1.69	●
	NCFI6R - 62.15 - 25x50 - YG812	NCFI06R6215025T	.244	.59	.004	.098	.197	.116	.71	1.69	●
	NCFI6R - 62.15 - 30x60 - YG812	NCFI06R6215030T	.244	.59	.004	.118	.236	.116	.71	1.69	●

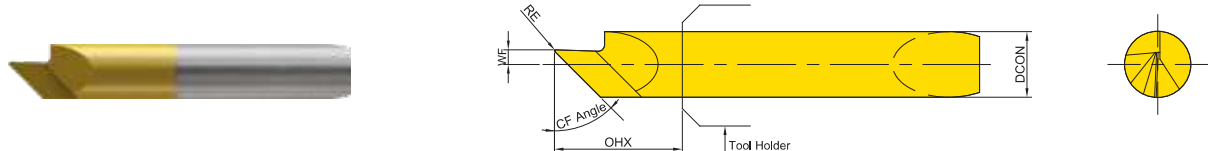
Boring Tool
FE - Face Grooving External



●: Stock item
○: Order made item

DCON (mm)	Designation	EDP	Inch ⁽ⁱⁿ⁾								YG812
			DMIN	LU	RE	CW	CDX	WF	OHX	OAL	
6 (.236)	NCFE6R - 62.15 - 10x20 - YG812	NCFE06R6215010T	.244	.59	.004	.039	.079	.116	.71	1.69	●
	NCFE6R - 62.15 - 15x30 - YG812	NCFE06R6215015T	.244	.59	.004	.059	.118	.116	.71	1.69	●
	NCFE6R - 62.15 - 20x40 - YG812	NCFE06R6215020T	.244	.59	.004	.079	.157	.116	.71	1.69	●
	NCFE6R - 62.15 - 25x50 - YG812	NCFE06R6215025T	.244	.59	.004	.098	.197	.116	.71	1.69	●
	NCFE6R - 62.15 - 30x60 - YG812	NCFE06R6215030T	.244	.59	.004	.118	.236	.116	.71	1.69	●

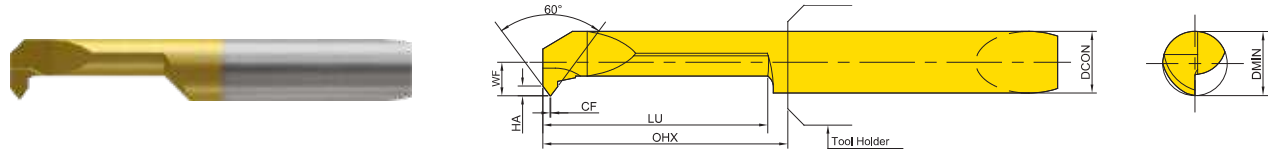
Boring Tool
CH - Chamfering



●: Stock item
○: Order made item

KAPR	DCON mm(inch)	Designation	EDP	Inch ⁽ⁱⁿ⁾						YG812
				DMIN	RE	CDX	WF	OHX	OAL	
45°	6 (.236)	NCCH6R - 10.03 - 45 - YG812	NCCH06R103045T	.039	.008	.14	.043	.51	1.50	○
60°		NCCH6R - 10.04 - 60 - YG812	NCCH06R103060T	.039	.008	.16	.020	.51	1.50	○

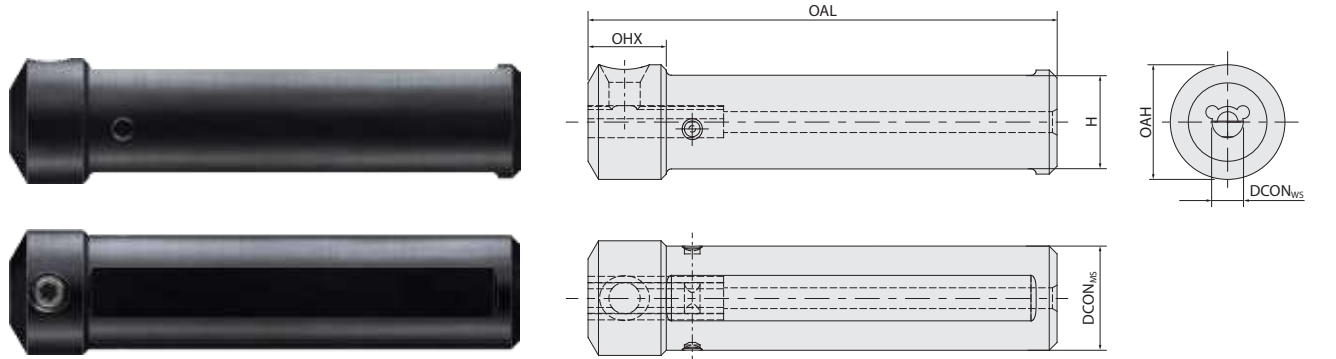
Boring Tool
TH - Threading



* P-60 : Partial Profile, 60°

DCON mm(inch)	Designation	EDP	Inch ⁽ⁱⁿ⁾									YG812
			DMIN	LU	TPIN	TPIX	WF	HA	CF	OHX	OAL	
4 (.157)	NCTH04R-42.15-050-P-60-YG812	NCTV04R4215050T	.165	.591	36.0	48.0	.079	.018	.002	.709	1.299	○
	NCTH04R-42.15-075-P-60-YG812	NCTV04R4215075T	.165	.591	24.0	36.0	.093	.026	.003	.709	1.299	○
	NCTH04R-42.15-100-P-60-YG812	NCTV04R4215100T	.165	.591	20.0	24.0	.089	.031	.004	.709	1.299	○
6 (.236)	NCTH06R-62.15-100-P-60-YG812	NCTV06R6215100T	.244	.591	20.0	24.0	.116	.031	.004	.807	1.693	○
	NCTH06R-62.15-125-P-60-YG812	NCTV06R6215125T	.244	.591	16.0	20.0	.091	.038	.005	.807	1.693	○
	NCTH06R-62.15-150-P-60-YG812	NCTV06R6215150T	.244	.591	14.0	16.0	.116	.045	.006	.807	1.693	○

Boring ToolHolder
Holder



Scale	Insert Size (DCON _{ws})	Designation	EDP	DCON _{MS}	OAL	H	OAH	OHX
Metric / Inch	4 (.157)	NCHI-0500.4i	ZBR0400500	.500	2.756	.421	.624	.551
		NCHI-0625.4i	ZBR0400625	.625	2.953	.546	.687	
		NCHI-0750.4i	ZBR0400750	.750	4.331	.671	.750	
	6 (.236)	NCHI-1000.4i	ZBR0401000	1.000	4.331	.906	1.000	
		NCHI-0500.6i	ZBR0600500	.500	2.756	.421	.663	
		NCHI-0625.6i	ZBR0600625	.625	2.953	.546	.726	
		NCHI-0750.6i	ZBR0600750	.750	4.331	.671	.827	
		NCHI-1000.6i	ZBR0601000	1.000	4.331	.906	1.000	

Infomation	Screw	Wrench
Description	-	TPWFTP15
EDP	TP156075	18000003